PE MAR-PE-13 Standard Operating Procedure for QoS / Congestion Event Resolution

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_9713

Description

QoS (Quality of Service) congestion events on Cisco PE MAR-PE-13, occurring at Marathahalli Core, are primarily attributed to microbursts, interface tail drops, and QoS queue starvation. This failure domain arises due to the complex interplay between the PE's traffic management capabilities, including the class-based queuing (CBQ) and weighted random early detection (WRED) algorithms. When a burst of high-priority traffic overwhelms the PE's QoS queues, it can lead to packet drops and subsequent congestion events, triggering ALARM_9713 on the Cisco PE. The sheer volume of traffic, combined with the PE's inability to manage it efficiently, results in a cascading effect, ultimately manifesting as a critical QoS congestion event.

Pre-requisite Checks

- Pre-requisite: Verify the redundant link status between MAR-PE-13 and its adjacent routers to ensure it is operational and not contributing to the congestion event.

- Pre-requisite: Check the physical environment of Marathahalli Core to ensure there are no cable or equipment issues that could be contributing to the failure.

- Pre-requisite: Confirm that the QoS policies in place are correctly configured and not causing the congestion event.

Potential Root Causes

Cause	Probability
Configuration mismatch between the QoS policies and the actual traffic flow.	High
Hardware limitations, such as insufficient buffer space or inadequate traffic handling capabilities of MAR-PE-13.	Medium
Software bugs or glitches affecting the PE's QoS management capabilities.	Low

Diagnostic Commands & Expected Outputs

```
Command: show policy-map interface GigabitEthernet0/0/0, Expected Output: Policy Map: , Description: This command displays the QoS policy map configuration applied to the GigabitEthernet0/0/0 interface, allowing us to verify the QoS policies in place and identify potential configuration mismatches.

Command: show queue interface GigabitEthernet0/0/0, Expected Output: Queue Statistics for GigabitEthernet0/0/0, Description: This command provides detailed queue statistics for the GigabitEthernet0/0/0 interface, enabling us to assess the current queue congestion levels and identify potential areas for improvement.
```

```
Command: show ip cache flow, Expected Output: Flow Cache Information, Description: This command displays the flow cache information, which can help identify the source and destination IP addresses, ports, and protocols involved in the congestion event.
```

Resolution Steps

Step 1: Step: Update the QoS policy map configuration to reflect the actual traffic flow., Command: configure terminal policy-map class bandwidth exit, Description: This command updates the QoS policy map configuration to ensure it correctly reflects the actual traffic flow, thereby preventing configuration mismatches and potential congestion events.

Step 2: Step: Adjust the queue settings on the GigabitEthernet0/0/0 interface to improve congestion handling., Command: interface GigabitEthernet0/0/0 queue-limit exit, Description: This command adjusts the queue settings on the GigabitEthernet0/0/0 interface to improve congestion handling, allowing the PE to better manage the traffic flow and prevent queue drops.

Step 3: Step: Clear the flow cache to prevent any stale information from affecting congestion event analysis., Command: clear ip cache flow, Description: This command clears the flow cache, ensuring that any stale information is removed and does not affect congestion event analysis.

Post-Check Verification

```
Command: show policy-map interface GigabitEthernet0/0/0, Expected Output: Policy Map: , Description: This command verifies that the updated QoS policy map configuration is correctly applied to the GigabitEthernet0/0/0 interface.

Command: show queue interface GigabitEthernet0/0/0, Expected Output: Queue Statistics for GigabitEthernet0/0/0 (Improved), Description: This command verifies that the updated queue settings on the GigabitEthernet0/0/0 interface have improved congestion handling and reduced queue drops.

Command: show ip cache flow, Expected Output: Flow Cache Information (Updated), Description: This command verifies that the flow cache information has been updated and does not contain any stale information.
```

Roll-back Procedures

- Step: Restore the original QoS policy map configuration., Command: configure terminal no policy-map policy-map exit, Description: This command restores the original QoS policy map configuration, ensuring that the PE returns to its previous state if the fix fails.

- Step: Reset the queue settings on the GigabitEthernet0/0/0 interface to their original values., Command: interface GigabitEthernet0/0/0 no queue-limit queue-limit exit, Description: This command resets the queue settings on the GigabitEthernet0/0/0 interface to their original values, allowing the PE to revert to its previous congestion handling behavior if the fix fails.

- Step: Re-populate the flow cache with the original information., Command: clear ip cache flow, Description: This command re-populates the flow cache with the original information, ensuring that the PE returns to its previous state if the fix fails.